Multimodal Computational Intelligence for Enhancing Caregiver Performance and Care Delivery – Edge-Enabled Skeletal Movement Analysis System for Real-Time Patient Fall Prediction

R26-DS-010

Project Proposal Report

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B.Sc. (Hons) Degree in Information Technology Specialized in Data Science

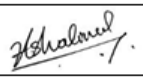
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DECLARATION

I declare that this is my own work, and this proposal does not knowingly incorporate any material previously submitted for a degree or diploma at any other university or higher education institution, nor does it contain any material that has been previously published or written by another person apart from where proper acknowledgment is given in the text.

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ABSTRACT

Patient falls remain a major safety concern in hospitals, nursing homes, and long-term care facilities, particularly among elderly patients and individuals with mobility limitations. Falls can lead to serious injuries such as fractures, head trauma, and prolonged hospitalization, placing significant pressure on healthcare systems and caregivers. Existing fall detection solutions often rely on wearable sensors or camera-based monitoring systems. However, wearable devices depend on patient compliance, while traditional video monitoring raises serious privacy concerns and may introduce latency when cloud-based processing is used. These limitations highlight the need for a reliable, privacy-preserving, and real-time monitoring solution that can assist caregivers in identifying potential fall risks.

This research proposes an Edge-Enabled Skeletal Movement Analysis System for Real-Time Patient Fall Prediction designed to monitor patient movement and predict potential fall events in real time. The system uses a standard USB camera to capture human movement and applies pose estimation techniques to extract skeletal joint coordinates while immediately discarding the original video frames to protect patient privacy. The extracted skeletal data is analysed using a Spatio-Temporal Graph Convolutional Network (ST-GCN) to understand spatial and temporal body movement patterns. By analysing motion trajectories, the system can differentiate between normal patient movements and high-risk fall behaviours, reducing false alarms and improving monitoring reliability.

The proposed system will be implemented using computer vision, Image Processing, Deep Learning, and Machine Learning techniques within an edge computing environment to enable low-latency processing without relying on cloud infrastructure. When a potential fall is detected, the system will generate real-time alerts for caregivers and provide a privacy preserving visualization of the movement event.

The expected outcome of this research is a practical and intelligent patient monitoring solution that improves fall detection accuracy, protects patient privacy, and enhances caregiver response time. The proposed system also has potential commercial applications in healthcare monitoring systems, smart hospitals, and assisted living environments.

Keywords: Fall Detection, Edge Computing, Spatio-Temporal Graph Convolutional Network (ST-GCN), Human Activity Recognition, Healthcare Monitoring System.

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LIST OF ABBREVIATIONS

Table 1: List of Abbreviations

Abbreviation	**Full Term**
CEAI	Centre of Excellence for Artificial Intelligence
AI	Artificial Intelligence
ML	Machine Learning
DL	Deep Learning
IP	Image Processing
DS	Data Science
ST-GCN	Spatio-Temporal Graph Convolutional Network
LSTM	Long Short-Term Memory
Bi-LSTM	Bidirectional Long Short-Term Memory
HAR	Human Activity Recognition
SDFA	Structure-Aware Discriminative Feature Aggregation
FR	Functional Requirement
NFR	Non-Functional Requirement

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1. INTRODUCTION

1.1. Background and Context

Patient falls remain one of the most common and serious safety issues in healthcare environments such as hospitals, nursing homes, and long-term care facilities. Many elderly or physically weak patients are at high risk of losing balance when attempting to move. Falls can lead to severe injuries including fractures, head trauma, and even death. According to global healthcare reports, millions of fall incidents occur each year among older adults, making fall prevention an important priority in modern healthcare systems. These incidents also increase healthcare costs, extend hospital stays, and create additional emotional stress for patients and caregivers. Therefore, developing intelligent monitoring systems that can detect and predict falls is essential for improving patient safety.

The population most affected by this issue includes elderly patients, individuals with mobility impairments, patients recovering from surgery, and those suffering from neurological conditions such as stroke or Parkinson's disease. In many hospitals and care homes, nurses or caregivers are responsible for monitoring multiple patients simultaneously, which makes continuous supervision difficult. As a result, falls may occur when medical staff are not immediately present. Early detection and timely alerts can significantly reduce the severity of injuries and allow caregivers to respond quickly during emergencies.

Several technological solutions have been introduced to address fall detection and monitoring. Traditional systems often use wearable sensors such as accelerometers and gyroscopes attached to the patient's body. While these devices can detect sudden movements, they rely heavily on patient compliance and may cause discomfort if worn for long periods. Additionally, elderly patients may forget to wear these devices or remove them unintentionally. Camera-based monitoring systems have also been developed to observe patient movement; however, these systems raise serious privacy concerns because they capture and store visual images of patients. Protecting patient privacy is a major challenge in healthcare monitoring technologies.

Recent advances in computer vision and deep learning have enabled the development of skeleton-based human activity recognition systems. These systems extract only the key joint coordinates of the human body instead of storing full video images. Spatial-Temporal Graph Convolutional Networks (ST-GCN) have been widely used for analyzing human movement patterns by modeling the body as a graph of connected joints over time [1]. This approach allows machines to understand complex human actions such as walking, sitting, or falling. Research has shown that combining pose estimation methods with graph-based neural networks can significantly improve the accuracy of fall detection systems [2], [4]. Furthermore, privacy-preserving monitoring methods that rely solely on skeleton data can protect patient identity while still enabling effective activity analysis [5].

Although these systems provide promising results, many existing solutions still depend on cloud computing or internet connectivity for processing video data. This dependency introduces network latency, increases operational costs, and raises additional privacy risks when sensitive data is transmitted over networks. In emergency situations such as falls, even a few seconds of delay can reduce the effectiveness of response actions. Therefore, there is a need for intelligent fall detection systems that can operate locally with minimal delay while preserving patient privacy.

To address these challenges, this research proposes an Edge-Enabled Skeletal Movement Analysis System for Real-Time Patient Fall Prediction for real-time patient monitoring. The proposed system uses a standard USB camera to capture movement data, which is processed directly on a local laptop to eliminate dependency on cloud servers and reduce latency. The system extracts human joint coordinates in real time to generate an anonymous stick-figure representation, while the original video frame is immediately discarded to ensure privacy. The extracted skeletal data is then analysed using an ST-GCN to understand body movement patterns and predict possible fall events. By analysing spatio-temporal motion trajectories, the system can differentiate between normal movements, such as adjusting position in bed, and dangerous fall behaviours, effectively reducing false alarms.

If a high-risk fall event is predicted, the system generates a multi-sensory alert notification—including audio and continuous vibration—on the caregiver's mobile device. Caregivers can then access a secure dashboard to view a short, privacy-preserving skeleton animation of the event, accompanied by a heatmap highlighting unstable joints. This approach combines realtime monitoring, predictive analytics, and privacy-aware design to create a safer healthcare environment.

This research aligns with United Nations Sustainable Development Goal (SDG) 3: Good Health and Well-being, by reducing preventable injuries, and SDG 9: Industry, Innovation, and Infrastructure, through the application of edge AI in healthcare. Furthermore, it aligns with the Centre of Excellence for AI (CEAI) research cluster, specifically within Image Processing (IP), Deep Learning (DL), and Machine Learning (ML). By integrating these at the edge, the project applies Data Science (DS) principles to function as an intelligent Decision Support System, translating real-time movement data into actionable, privacy-safe alerts.

From an academic perspective, this research aligns directly with the focus areas of the Centre of Excellence for AI (CEAI). Specifically, the system's reliance on local camera feeds to extract joint coordinates is deeply rooted in Image Processing (IP), while the ST-GCN leverages advanced Deep Learning (DL) and Machine Learning (ML). By integrating these techniques at the edge, the project applies core Data Science (DS) principles to function as an intelligent Decision support system, effectively translating complex, real-time human movement data into actionable, privacy-safe alerts for healthcare environments.

1.2 Literature Review

Recent research on patient monitoring and fall detection has increasingly focused on using computer vision and deep learning techniques to identify risky human movements in real time. One important direction in this area is skeleton-based action recognition, where the human body is represented as a set of joints instead of raw video frames. The work by Yan *et al.* introduced the Spatio-Temporal Graph Convolutional Network (ST-GCN), which became a foundational method for analyzing human actions from skeleton sequences [1]. This approach models body joints as graph nodes and captures both spatial and temporal relationships, making it highly suitable for movement understanding tasks such as fall detection. Since the proposed research also relies on skeleton data and graph-based modeling, this study provides the core technical background for the project.

Several recent studies have applied pose estimation and graph-based learning directly to fall detection. The study on a real-time fall detection and reporting system using AlphaPose demonstrates how body key points can be extracted from video and then used to identify fall events in a more efficient and automated way [2]. Similarly, research on impact detection in fall events using spatio-temporal graph convolutional networks further shows that graph-based methods can effectively detect dangerous fall-related movement patterns from 3D skeleton data [4]. These studies are highly relevant because they confirm that skeleton-based analysis can be used for real-time healthcare monitoring. However, many existing studies mainly focus on identifying a fall after it happens, rather than predicting a high-risk event before the fall is fully completed.

In addition to fall detection, some research has explored human motion prediction, which is important for building systems that can identify dangerous movement trends before injury occurs. The BeLFusion model presents a behaviour-driven motion prediction approach using latent diffusion techniques [3]. This work is useful because it shows that future body positions can be estimated from previous skeletal movement patterns. For the proposed research, this idea is important because the system aims not only to recognize movement, but also to predict whether the patient is moving into an unstable state. However, advanced motion prediction models such as diffusion-based frameworks may require greater computational resources and may be difficult to deploy efficiently on low-power edge devices.

Privacy is another major concern in vision-based healthcare systems. Traditional camera monitoring can raise ethical and practical issues because patients may feel uncomfortable being recorded continuously. In response to this challenge, the study on a skeleton-based privacypreserving smart activity sensor proposes using skeletal representations instead of raw video images for senior care and patient monitoring [5]. This approach is closely aligned with the proposed research, which also intends to discard the original video after extracting joint coordinates. By preserving only anonymous skeletal data, such systems can reduce privacy risks while still supporting movement analysis. However, privacy-preserving approaches may still face limitations when skeletal extraction quality decreases due to poor lighting, occlusion, or unusual body postures.

Recent literature has also shown growing interest in improving fall detection performance in real-time edge environments. The study on a real-time fall detection model based on BlazePose and improved ST-GCN presents an efficient method for combining lightweight pose estimation

with graph-based movement analysis [6]. This is especially relevant to the proposed research because the system is designed to operate fully on a local laptop without cloud support. Likewise, the study on a deep learning and edge computing integrated approach for fall detection highlights the importance of performing fall analysis close to the data source in order to reduce latency and improve response time [8]. These works support the use of edge AI in healthcare applications. Even so, edge-based systems still face practical limitations such as hardware constraints, lower processing power, and challenges in maintaining high accuracy under real-world conditions.

Other studies have attempted to improve skeleton-based fall detection by modifying the graph learning process or by combining multiple temporal learning strategies. For example, research on skeleton-based fall detection using spatial-temporal graph convolutional networks with learnable edges suggests that adaptive graph structures can improve action understanding by learning more meaningful joint relationships [7]. In a similar way, the study on multimodal fall detection using spatial-temporal attention and Bi-LSTM-based feature fusion explores attention mechanisms and recurrent learning for better sequence modelling [9]. The SDFA model also contributes to efficient human fall detection by focusing on structure-aware feature aggregation from video [10]. These approaches demonstrate that researchers are actively trying to improve model accuracy, robustness, and efficiency. However, many of these systems still concentrate mainly on classification accuracy and do not fully address the combined need for privacy preservation, low false alarm rates, local edge deployment, and clear caregiver-facing visual explanation.

Beyond research studies, several existing products and services in the healthcare market use wearable devices, panic buttons, smart watches, pressure mats, and CCTV-based monitoring to support patient safety. Wearable systems can be useful for detecting sudden impact, but they depend heavily on patient compliance and may be uncomfortable for elderly users. Bedside alarms and pressure sensors are simple to deploy, yet they often trigger alerts only after a patient has already attempted to leave the bed, which may be too late in some cases. Camera-based monitoring systems can provide richer information, but they introduce privacy concerns and may require cloud connectivity or large storage capacity. Therefore, although existing products offer partial solutions, they often do not provide a balanced combination of real-time prediction, privacy preservation, false alarm reduction, and edge-based operation.

Overall, the existing literature shows strong progress in skeleton-based activity recognition, fall detection, motion prediction, and privacy-aware healthcare monitoring [1]–[10]. However, there is still a need for an integrated system that can process video locally on-device, remove identifiable visual data immediately, analyse skeletal movement using ST-GCN-based techniques, reduce unnecessary alarms caused by safe body adjustments, and provide caregivers with a privacy-safe visual explanation of the detected event. This gap in the literature supports the motivation for the proposed research.

Comparison With Existing Fall Detection Systems

Table 2 : Comparison with Existing Fall Detection

Existing Approaches	Proposed System
Many traditional fall detection systems rely on wearable sensors such as accelerometers or smart devices attached to the patient's body. These systems depend on patient compliance, and elderly patients may forget or refuse to wear the device.	The proposed system uses a camera-based skeletal analysis approach, eliminating the need for wearable devices while continuously monitoring patient movement.
Conventional camera-based monitoring systems store full video footage, which raises serious privacy concerns in healthcare environments.	The proposed system extracts skeletal joint coordinates using MediaPipe Pose and immediately deletes the original video frames, ensuring privacy-preserving monitoring.
Many existing systems detect falls only after they occur, limiting the ability to prevent injuries.	The proposed system analyzes spatiotemporal skeletal movement patterns using Spatio-Temporal Graph Convolutional Networks (ST-GCN) to predict high-risk fall movements before the fall occurs.
Some monitoring systems rely on cloudbased processing, which introduces network latency and potential data security risks.	The proposed system performs edge-based processing on a local device, reducing latency and keeping sensitive patient data within the local environment.
Existing systems often produce high false alarm rates, especially when patients perform normal movements such as adjusting in bed.	The proposed system applies motion trajectory analysis and intelligent filtering to distinguish between normal adjustments and dangerous fall patterns, reducing false alarms.

1.3 Research Gap

Although significant progress has been made in fall detection and human activity recognition, several important limitations still exist in current systems. Many existing solutions rely heavily on wearable sensors such as accelerometers or smart devices attached to the body. While these systems can detect sudden movement changes, they depend on the patient consistently wearing the device, which may not always be practical for elderly individuals or patients with cognitive impairments. Additionally, wearable sensors may fail to provide detailed information about body posture or movement patterns, which limits their ability to accurately identify complex fall dynamics.

Vision-based monitoring systems have also been proposed as an alternative to wearable solutions. These systems typically use computer vision and deep learning techniques to analyse human movement captured by cameras. For example, Spatio-Temporal Graph Convolutional Networks (ST-GCN) have been widely used to model skeletal joint movements and recognize human activities such as walking, sitting, and falling [1]. Similarly, pose estimation methods such as AlphaPose have been applied to extract body key points from video frames for fall detection applications [2]. While these approaches improve the ability to recognize fall events, many of them focus mainly on detecting a fall after it occurs rather than predicting high-risk movements before the fall happens.

Another limitation in existing research is the reliance on cloud-based processing architectures. Many camera-based monitoring systems transmit video data to remote servers for analysis, which introduces additional network latency and raises privacy concerns when sensitive patient information is transmitted over external networks. In healthcare environments where immediate response is critical, even a short delay in processing may reduce the effectiveness of emergency interventions. Although recent studies have explored edge computing for fall detection to reduce latency, these systems still face challenges related to computational efficiency and real-time performance on local devices [6], [8].

Privacy protection is another major challenge in current monitoring technologies. Traditional camera surveillance systems capture and store full video footage of patients, which can expose personal and sensitive information. Recent research has proposed skeleton-based monitoring approaches that store only joint coordinate data rather than full images in order to improve privacy protection [5]. However, many of these approaches still retain portions of visual data or require additional processing stages that may compromise real-time performance. Furthermore, existing systems rarely provide clear visual explanations of detected fall events that caregivers can easily interpret during emergency situations.

Another limitation observed in current literature is the high rate of false alarms generated by many fall detection systems. Many algorithms classify any sudden movement or body posture change as a potential fall, even when the patient is performing normal actions such as turning in bed or adjusting their sitting position. Although several advanced models, including attention-based networks and Bidirectional Long Short-Term Memory (Bi-LSTM) models, have attempted to improve temporal sequence analysis for fall detection [9], these systems still struggle to effectively distinguish between safe movements and dangerous fall dynamics. As a result, frequent false alarms may reduce caregiver trust in automated monitoring systems.

Furthermore, most existing studies focus primarily on improving model accuracy, often overlooking the need for an integrated system that simultaneously addresses privacy protection, real-time processing, predictive analysis, and user-friendly visualization for caregivers. While advanced models such as Structure-Aware Discriminative Feature Aggregation (SDFA) have improved feature extraction for fall detection tasks [10], they are typically evaluated under controlled experimental conditions and may not fully address the practical challenges faced in real healthcare environments.

Therefore, there remains a clear research gap in developing a privacy-preserving, edge-enabled fall prediction system that can analyse skeletal movement data in real time while minimizing false alarms and protecting patient identity. The proposed research aims to address this gap by introducing an Edge-Enabled Skeletal Movement Analysis System that processes video locally using a standard camera and extracts only skeletal joint coordinates through image processing techniques. The system then applies ST-GCN-based movement analysis to monitor spatiotemporal body dynamics and predict potential fall events before they fully occur.

The novelty of this research lies in the integration of several key features into a single intelligent monitoring framework. First, the system operates entirely on a local edge device, eliminating the need for cloud-based processing and reducing latency. Second, it ensures patient privacy by immediately discarding raw video frames and retaining only anonymous skeletal representations. Third, the system incorporates predictive movement analysis to distinguish between normal body adjustments and dangerous fall patterns, thereby reducing unnecessary alerts. Finally, the proposed system provides caregivers with a privacy-safe visual explanation of detected events through a short skeleton replay with instability heatmap visualization.

By combining edge computing, skeleton-based human activity recognition, and graph-based deep learning, this research contributes a novel approach for intelligent patient monitoring systems. The proposed solution aims to address key limitations identified in existing studies [1]– [10] and provide a practical, privacy-aware, and real-time fall prediction system suitable for deployment in healthcare environments.

2. OBJECTIVE

2.1 Main Objective

To design, develop, and evaluate an edge-enabled skeleton-based system for real-time patient fall detection and prediction by extracting skeletal joint coordinates using pose estimation techniques and analysing spatio-temporal movement patterns with a Spatio-Temporal Graph Convolutional Network (ST-GCN), in order to achieve high detection accuracy, low latency processing, and reduced false alarm rates within a privacy-preserving healthcare monitoring environment [1], [4], [8].

2.2 Specific Objectives

The following objectives focus on the development of the Edge-Enabled Skeletal Movement Analysis System for Real-Time Patient Fall Prediction component of the proposed system.

SO1 – To develop a skeletal joint extraction module using real-time pose estimation techniques.

This objective focuses on capturing patient movement through a camera and extracting skeletal joint coordinates using IP methods. Pose estimation frameworks such as MediaPipe Pose can detect body landmarks and generate skeletal representations from video frames, enabling privacy-preserving human activity analysis.

SO2 – To implement a ST-GCN model for analysing skeletal movement patterns.

This objective aims to apply DL and ML techniques to model spatial and temporal relationships between body joints for movement analysis. ST-GCN has been widely used for skeleton-based HAR tasks and movement classification [1].

SO3 – To design a motion trajectory analysis mechanism to identify high-risk fall movement patterns.

This objective focuses on analysing sequential skeletal data to distinguish between normal body adjustments and potentially dangerous fall behaviours. Temporal analysis of joint movement patterns helps improve fall detection reliability and reduce false alarms [4].

SO4 – To integrate the skeletal movement analysis module with an edge-based processing system.

This objective ensures that all movement analysis tasks are performed locally on an edge device without relying on cloud infrastructure. Edge computing architectures have been explored in fall detection systems to reduce latency and improve real-time performance [8].

SO5 – To evaluate the performance of the skeletal movement analysis model using accuracy, latency, and false alarm rate metrics.

These objective measures the effectiveness of the proposed system in detecting fall-related movement patterns under real-time monitoring conditions using standard evaluation metrics used in fall detection research [7], [10].

3. METHODOLOGY

This research adopts a Design Science Research (DSR) approach to design, develop, and evaluate an intelligent system that addresses the real-world problem of patient fall detection in healthcare environments. Design Science Research focuses on creating innovative technological solutions and evaluating their effectiveness in solving practical problems. In this study, the developed artifact is an Edge-Enabled Skeletal Movement Analysis System for RealTime Patient Fall Prediction that predicts potential patient falls using real-time skeletal movement data.

The research process involves designing a system that captures human movement through a camera, converts visual frames into skeletal joint coordinates using Image Processing (IP) techniques, and analyses the resulting motion patterns using Deep Learning (DL) algorithms. The system will then be experimentally evaluated to determine its effectiveness in detecting or predicting fall events while reducing false alarms. This approach allows the research to integrate system development, machine learning modelling, and performance evaluation within a realistic healthcare monitoring context.

The proposed system consists of multiple processing stages that work together to analyse patient movement and identify potential fall risks.

First, skeletal joint coordinates will be extracted from video frames using MediaPipe Pose (BlazePose), a real-time pose estimation framework developed for lightweight computer vision applications. MediaPipe Pose detects key body landmarks such as shoulders, hips, knees, and ankles from each frame and represents the human body as a skeletal structure. This approach allows the system to analyse human movement without storing full video images, which helps protect patient privacy.

Next, the extracted skeletal sequences will be analysed using a Spatio-Temporal Graph Convolutional Network (ST-GCN). In this model, body joints are treated as nodes in a graph while bones represent edges connecting these nodes. The ST-GCN captures both spatial relationships between joints and temporal changes in movement across consecutive frames. This approach has been widely used for skeleton-based Human Activity Recognition (HAR) and fall detection tasks due to its ability to model complex body dynamics [1], [4].

To reduce false alarms, the system will implement a temporal motion analysis component that examines the trajectory and velocity of joint movements across time. By analysing motion patterns within short time windows, the system can distinguish between normal body adjustments—such as repositioning in bed—and high-risk movements associated with falls. Similar temporal modelling strategies have been explored in previous fall detection studies to improve prediction reliability [9].

Finally, when the system detects a potential fall event, it will generate a notification alert for caregivers. The alert system will trigger mobile notifications and provide a short skeletal animation replay with an instability heatmap highlighting joints involved in the fall movement.

The system will be implemented using several software tools and platforms that support machine learning, computer vision, and real-time data processing.

The primary programming language used will be Python, which provides extensive support for data science and machine learning development. Pose estimation will be implemented using MediaPipe, a framework developed by Google for efficient real-time body landmark detection.

For deep learning model implementation, frameworks such as PyTorch or TensorFlow will be used to develop and train the ST-GCN model. These frameworks provide optimized libraries for building neural networks and processing large-scale sequential data.

The system will run on a local edge device, specifically a laptop connected to a standard USB camera. Processing the data locally eliminates reliance on cloud servers and significantly reduces network latency. This edge-based architecture improves real-time performance while ensuring that sensitive patient information remains within the local device environment.

Additionally, a lightweight monitoring dashboard will be developed to allow caregivers to view alerts and review skeleton-based movement visualizations when fall events are detected.

The system will be trained and evaluated using publicly available skeleton-based datasets related to Human Activity Recognition (HAR) and fall detection. These datasets contain annotated skeletal sequences representing different human movements, including normal daily activities and fall events.

To simulate real-world conditions, additional controlled movement recordings may be captured using a camera in a safe and controlled environment. These recordings will help test the system's ability to distinguish between normal patient movements and fall-related behaviour.

Since the research involves analysing human movement data, ethical considerations related to privacy and data protection are extremely important. The proposed system addresses these concerns by immediately discarding raw video frames after extracting skeletal joint coordinates. Only anonymous skeletal data will be processed and stored, ensuring that no identifiable images of patients are retained. This privacy-preserving monitoring approach has been recommended in recent healthcare research studies [5].

All data collection and system testing procedures will follow ethical guidelines related to responsible artificial intelligence development and data privacy protection.

The effectiveness of the proposed system will be evaluated through experimental testing and quantitative performance analysis. The system will be tested under different movement scenarios including normal patient activities and simulated fall events.

Several evaluation metrics will be used to assess the model's performance. These include accuracy, precision, recall, and F1-score, which measure the ability of the system to correctly detect fall events while minimizing incorrect predictions. These metrics are commonly used in machine learning-based fall detection research [7], [10].

In addition to classification performance, the system's response time (latency) will also be evaluated to determine how quickly the system can generate alerts after detecting risky movement patterns. Low latency is essential in healthcare monitoring environments where rapid response can reduce injury risks.

Another important metric is the false alarm rate, which measures how often the system incorrectly identifies normal movements as fall events. Reducing false alarms is crucial to maintain caregiver trust and ensure that the monitoring system remains reliable in real-world healthcare settings.

Through these validation strategies, the research aims to demonstrate that the proposed EdgeEnabled Skeletal Movement Analysis System for Real-Time Patient Fall Prediction can provide accurate, privacy-preserving, and real-time fall prediction capabilities suitable for healthcare environments.

4. HIGH-LEVEL SYSTEM ARCHITECTURE

4. 1. Overall System Architecture

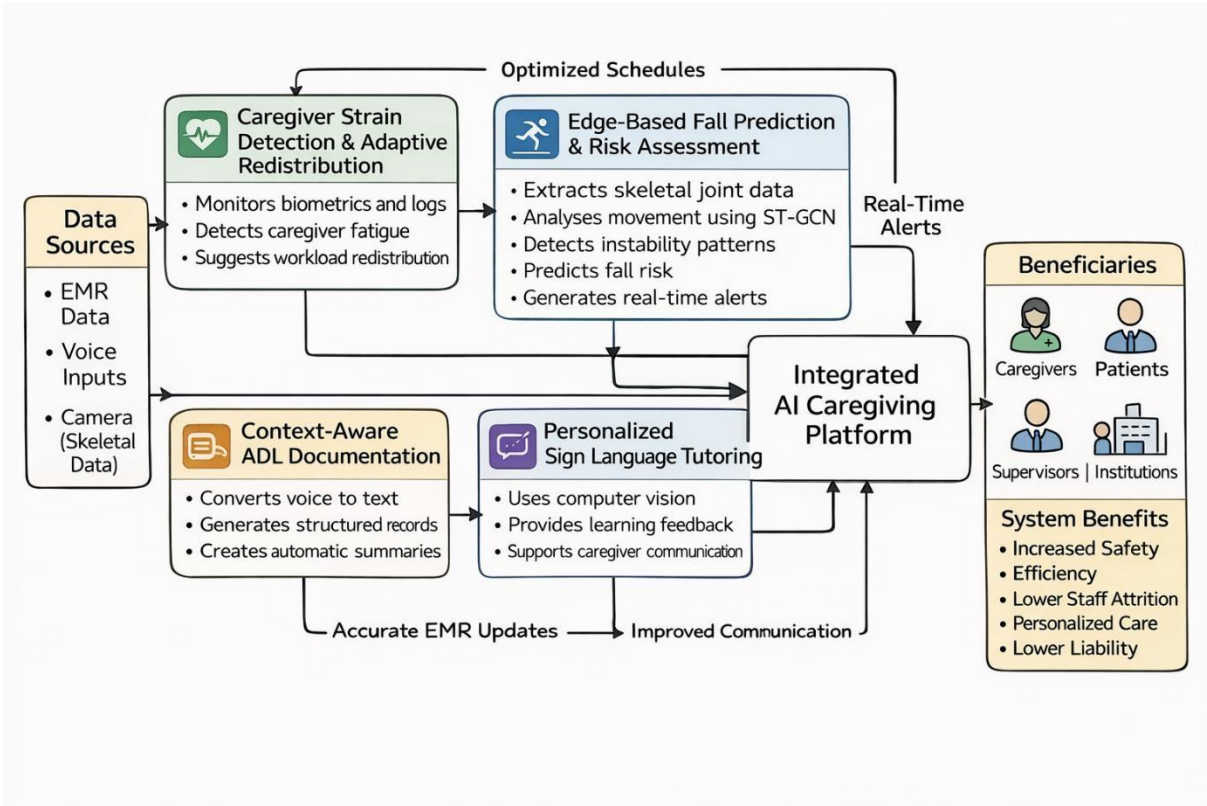


Figure 1: Overview of the integrated AI-based caregiving system showing data flow and component interactions.

4. 2. Component Overview

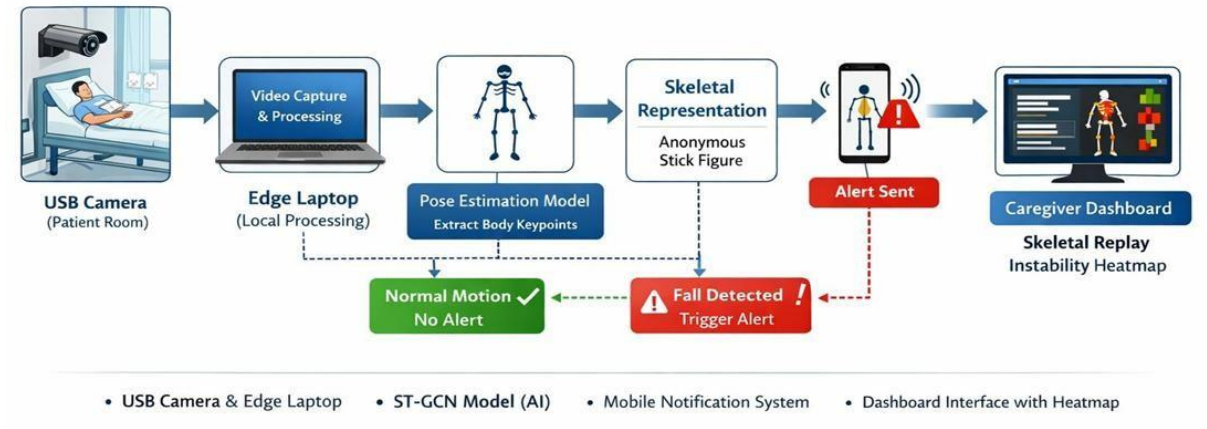


Figure 2 : High-level system architecture of the proposed edge-enabled skeletal movement analysis component.

5. USER REQUIREMENTS

The proposed Edge-Enabled Skeletal Movement Analysis System for Real-Time Patient Fall Prediction is intended to support several stakeholders involved in healthcare monitoring and patient safety.

- Patients are the primary stakeholders of the system. Elderly individuals, patients with mobility impairments, and those recovering from medical conditions are particularly vulnerable to falls. The system aims to provide continuous monitoring of patient movement while preserving privacy by analysing skeletal joint coordinates instead of storing visual images.
- Caregivers are the main users of the monitoring system. They are responsible for supervising multiple patients simultaneously and responding quickly when emergency situations occur. The proposed system assists caregivers by automatically detecting potential fall events and sending alert notifications to allow faster intervention.
- Healthcare administrators are also stakeholders because they are responsible for ensuring patient safety and improving hospital operational efficiency. A reliable automated monitoring system can reduce the risk of patient injuries and help healthcare institutions improve safety standards.

Finally, system administrators and developers are responsible for maintaining the monitoring platform, ensuring that the AI models, DL algorithms, and IP components function correctly within the edge computing environment.

5.1 Functional Requirements

The functional requirements describe the core system features needed to support fall detection and monitoring.

1. **FR1** – Real-Time Movement Capture
   The system shall capture live video data from a USB camera installed in the patient's room to monitor patient movements continuously.
2. **FR2** – Skeletal Joint Extraction
   The system shall extract skeletal joint coordinates from each video frame using pose estimation techniques such as MediaPipe Pose through IP methods.
3. **FR3** – Privacy-Preserving Data Processing
   The system shall immediately discard raw video frames after extracting skeletal joint coordinates to ensure that patient identity is not stored or transmitted.
4. **FR4** – Skeletal Movement Analysis
   The system shall analyse the extracted skeletal data using a Spatio-Temporal Graph Convolutional Network (ST-GCN) model to recognize human activity patterns and detect fall-related movements [1].
5. **FR5** – Fall Risk Prediction
   The system shall analyse spatio-temporal motion trajectories to identify high-risk movement patterns that may lead to a fall before the event occurs.

6. **FR6** – False Alarm Filtering
   The system shall differentiate between normal body movements (e.g., turning in bed or sitting adjustments) and dangerous fall patterns to reduce false alerts.
7. **FR7** – Alert Notification Generation
   The system shall generate a multi-sensory alert notification, including audio and vibration alerts, on the caregiver's mobile device when a potential fall event is detected.
8. **FR8** – Event Visualization Dashboard
   The system shall provide caregivers with a dashboard that displays a short skeletonbased replay of the detected movement event along with a heatmap indicating unstable joints.
9. **FR9** – System Monitoring and Control
   The system shall allow authorized users to access system status, monitor alerts, and review past event logs through a secure interface.

5.2 Non-Functional Requirements

Non-functional requirements define system quality attributes such as performance, security, reliability, and usability.

1. Performance Requirements
   - **NFR1** – Real-Time Processing Speed
     The system shall process incoming video frames and update skeletal movement analysis within 200 milliseconds per frame to maintain real-time monitoring capability.
   - **NFR2** – Alert Response Time
     The system shall generate a caregiver alert notification within 2 seconds after detecting a high-risk fall event.
   - **NFR3** – System Throughput
     The system shall process at least 20–30 frames per second (FPS) to ensure smooth real-time movement tracking.

2. Security Requirements
   - **NFR4** – Data Privacy Protection
     The system shall ensure that no raw video footage containing identifiable patient information is stored after skeletal joint extraction.
   - **NFR5** – Secure Access Control
     Only authorized caregivers or administrators shall be allowed to access the monitoring dashboard through authenticated login credentials.
   - **NFR6** – Secure Data Transmission
     All alert notifications and dashboard communications shall be transmitted through encrypted communication protocols to protect sensitive healthcare data.

3. Reliability Requirements
    - **NFR7** – System Availability
      The monitoring system shall maintain an operational availability of at least 99% uptime to ensure continuous patient monitoring.
    - **NFR8** – False Alarm Rate
      The system shall maintain a false alarm rate of less than 10% during fall detection experiments to ensure caregiver trust.

4. Usability Requirements
    - **NFR9** – User Interface Simplicity
      The caregiver dashboard shall allow users to view alert notifications and event replays within three interaction steps or less.
    - **NFR10** – Alert Visibility
      Alert notifications shall include visual, vibration, and audio indicators to ensure that caregivers can notice alerts even in busy hospital environments.
5. Scalability Requirements
    - **NFR11** – Edge Processing Capability
      The system shall operate entirely on a local edge device without requiring cloud-based processing, ensuring scalability and low-latency performance.

6. COMMERCIALIZATION PLAN

The proposed Edge-Enabled Skeletal Movement Analysis System can be commercialized as part of a larger intelligent healthcare support platform targeting healthcare institutions and caregiving services. The primary target market includes hospitals, elderly care homes, rehabilitation centers, and home healthcare providers, as these environments frequently deal with patient fall risks and require continuous monitoring solutions. The key customer personas include hospital administrators, nursing home managers, and care supervisors who are responsible for ensuring patient safety and operational efficiency.

The value proposition of the system lies in its ability to provide real-time fall detection and prediction while preserving patient privacy through skeleton-based analysis. Unlike traditional camera-based systems, the proposed solution does not store raw video data, which addresses privacy concerns. Additionally, the system reduces caregiver workload by minimizing the need for constant manual supervision and enables faster response during emergency situations. As highlighted in the presentation, improving patient safety and reducing caregiver burden are key benefits that increase the system's practical value in healthcare environments.

The revenue model can be designed as a subscription-based Software-as-a-Service (SaaS) model combined with hardware integration. Healthcare institutions can subscribe to the system on a monthly or annual basis depending on the number of monitored rooms or patients. Additional revenue can be generated through enterprise licensing, system customization, and integration with existing hospital management systems. For smaller care centers, a low-cost deployment model using edge devices such as laptops or Raspberry Pi can be introduced to improve accessibility.

In terms of cost estimation, the system is designed to be cost-effective by using commonly available hardware such as USB cameras and edge devices. The major costs include hardware setup (camera and edge processing unit), software development, model training, and system maintenance. Compared to traditional surveillance systems that require cloud infrastructure and large storage capacity, this approach significantly reduces operational costs by performing all processing locally.

The pricing strategy will focus on affordability and scalability. A tiered pricing model can be introduced, where small care centers pay a lower fee for basic monitoring features, while hospitals and large institutions can opt for advanced features such as analytics dashboards and multi-room monitoring. This flexible pricing approach ensures that the system can be adopted by a wide range of healthcare providers.

The proposed system has a strong competitive advantage due to its combination of edge computing, privacy-preserving skeleton analysis, and real-time fall prediction. Unlike existing solutions that either rely on wearable devices or cloud-based video monitoring, this system operates independently of internet connectivity and eliminates privacy risks associated with storing video data. Additionally, the ability to predict falls before they occur and provide explainable visual feedback gives it a significant advantage over traditional fall detection systems.

Finally, intellectual property considerations include protecting the system's core algorithms, model architecture, and system design. The novelty of combining edge-based processing with skeleton-based fall prediction can be protected through software copyrights and potential patents. Additionally, proprietary datasets and trained models can be considered valuable intellectual assets. As indicated in the presentation, intellectual property rights such as algorithm innovation, system architecture design, and proprietary datasets play an important role in maintaining a unique competitive position in the healthcare technology market.

7. BUDGET AND JUSTIFICATION

Estimated Costs for Research Project

Description	Hours / Units	Cost (LKR)
Labour Costs		
Software Development	350 hrs	42,000
System Integration	250 hrs	30,000
Testing & Deployment	200 hrs	24,000
Total Labour Cost		**96,000**
Material & Infrastructure Costs		
USB Camera for Fall Detection	1 unit	12,000
Cloud Hosting & Storage (6 months)	–	10,000
Software Tools / APIs / GPU Usage (4 months)	–	8,000
Total Material & Infrastructure Cost		**30,000**
GRAND TOTAL		**126,000**

Figure 3: Budget Table

The estimated budget for the proposed Edge-Enabled Skeletal Movement Analysis System includes hardware, software, cloud services, and human effort costs. The total estimated cost for the research project is approximately LKR 126,000, which is considered cost-effective for developing a real-time AI-based healthcare monitoring solution.

The primary hardware requirement for the system includes a USB camera and an edge processing device. A standard USB camera is estimated at approximately LKR 12,000, which is sufficient for capturing real-time patient movement. The edge device is assumed to be already available for development purposes; however, if required, an additional cost may be incurred. These hardware components are essential for enabling real-time data capture and local processing without relying on external infrastructure.

The proposed system mainly uses open-source frameworks such as MediaPipe, TensorFlow, and PyTorch, which reduces licensing costs. However, a small allocation of approximately LKR 8,000 is considered for optional software tools, APIs, or GPU usage required for model training and experimentation. This ensures flexibility in development and supports efficient model performance.

Although the system is designed to operate primarily on edge devices, limited cloud services may be required for data backup, testing, or remote access during development. An estimated cost of LKR 10,000 is allocated for cloud hosting and storage over a period of six months. This cost is relatively low compared to traditional cloud-dependent systems, as the proposed solution minimizes cloud usage.

Human effort represents the largest portion of the budget, covering development, integration, testing, and deployment activities. Software development is estimated at 350 hours (LKR

42,000), system integration at 250 hours (LKR 30,000), and testing and deployment at 200 hours (LKR 24,000). The total labour cost is therefore approximately LKR 96,000. These estimates reflect the time required to design, implement, and evaluate the system within the project timeline.

Each cost component is justified based on its contribution to the system. Hardware costs are necessary for real-time data acquisition and edge-based processing. Software-related costs support the development and training of machine learning models. Cloud services, although minimal, provide additional support for testing and data management. The human effort cost is essential as the project involves complex tasks such as pose estimation, graph-based deep learning, system integration, and real-time performance evaluation. Overall, the budget is optimized to balance cost efficiency with system performance, making the solution practical for real-world healthcare deployment.

8. WORK BREAKDOWN STRUCTURE (WBS)

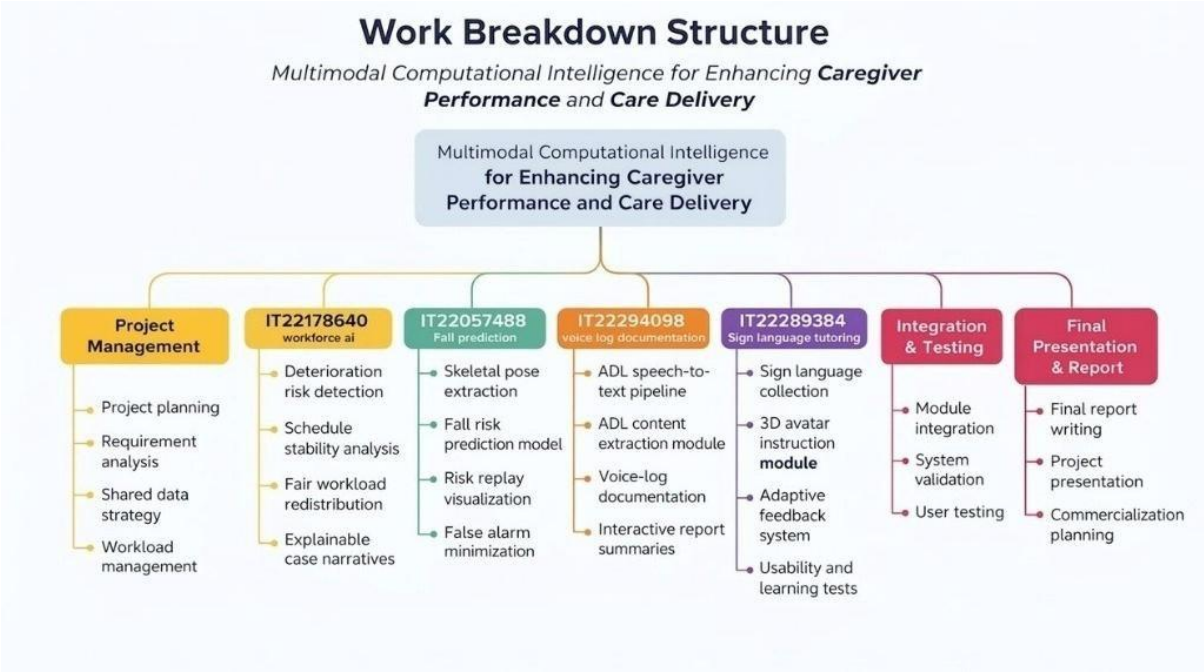


Figure 4: Work Breakdown Structure (WBS) of the proposed AI-based caregiving system.

Task Breakdown per Member • Each team member is assigned a clearly defined module based on project requirements and individual expertise.

- Responsibilities are distributed across project management, workforce AI, fall prediction, voice-log documentation, sign language tutoring, integration, and final reporting.
- This ensures accountability, specialization, and smooth progress tracking.

Timeline Alignment

- Tasks are scheduled in a phased manner so that each module progresses according to the overall project timeline.
- Dependencies between modules are considered to ensure timely integration, testing, and final delivery.
- Regular milestones help monitor completion and keep the project on schedule.

Balanced Workload Distribution

- Work is allocated evenly among members to avoid overloading any individual.
- Task complexity, effort required, and deadlines are considered while assigning responsibilities.
- This promotes efficient collaboration, fair contribution, and better project outcomes.

9. GANTT CHART

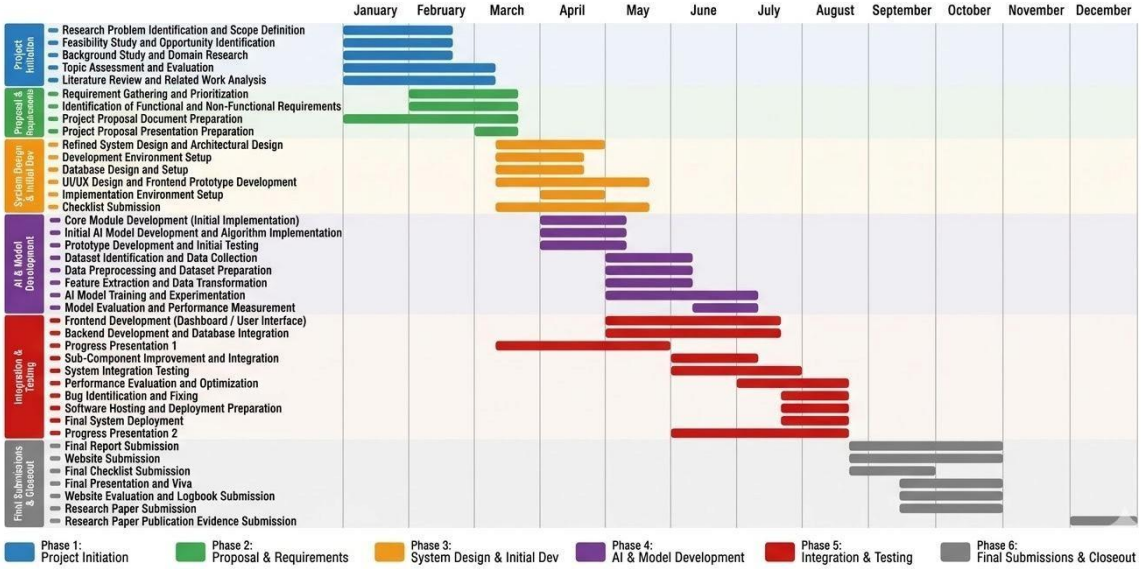


Figure 5:Gantt chart illustrating the project timeline and phase-wise task distribution.

10. AI USAGE DISCLOSURE

AI tools were used in a limited and supportive capacity during the preparation of this research proposal. ChatGPT and Google Gemini was utilized primarily to enhance language clarity, refine sentence structure, and assist in the organization of ideas. The purpose of using these tools was to improve the readability and presentation of the document, while ensuring that all core research concepts, problem formulation, and methodological decisions were developed independently by the author.

The extent of AI usage was strictly limited to minor writing assistance and idea refinement and did not involve generating full sections of the proposal or replacing original intellectual contribution. All AI-assisted outputs were critically reviewed, modified, and validated by the author to ensure accuracy, relevance, and alignment with academic standards. Validation methods included cross-checking information with credible academic sources, verifying technical correctness, and ensuring consistency with the research objectives. Full responsibility for the content presented in this proposal is retained by the author.

I hereby declare that:

1. I have read and understood the AI Usage Policy for the Research Project (IT4010).
2. I acknowledge that AI tools are to be used only as support tools, not as substitutes for my own understanding and work.
3. I agree to use AI tools responsibly, ethically, and transparently.
4. I will clearly disclose any use of AI tools in my submissions.
5. I understand that I bear full responsibility and accountability for any content, code, model, or output generated using AI tools.
6. I confirm that I will be able to explain, justify, and defend all aspects of my work during evaluations.
7. I understand that failure to comply with this policy may result in academic penalties.

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- Student ID: IT22057488
- Project Title: Edge-Enabled Skeletal Movement Analysis System for Real-Time Patient Fall Prediction
- Group ID : R26-DS-010

Signature

- Student Signature: 
- Date: 23.03.2026

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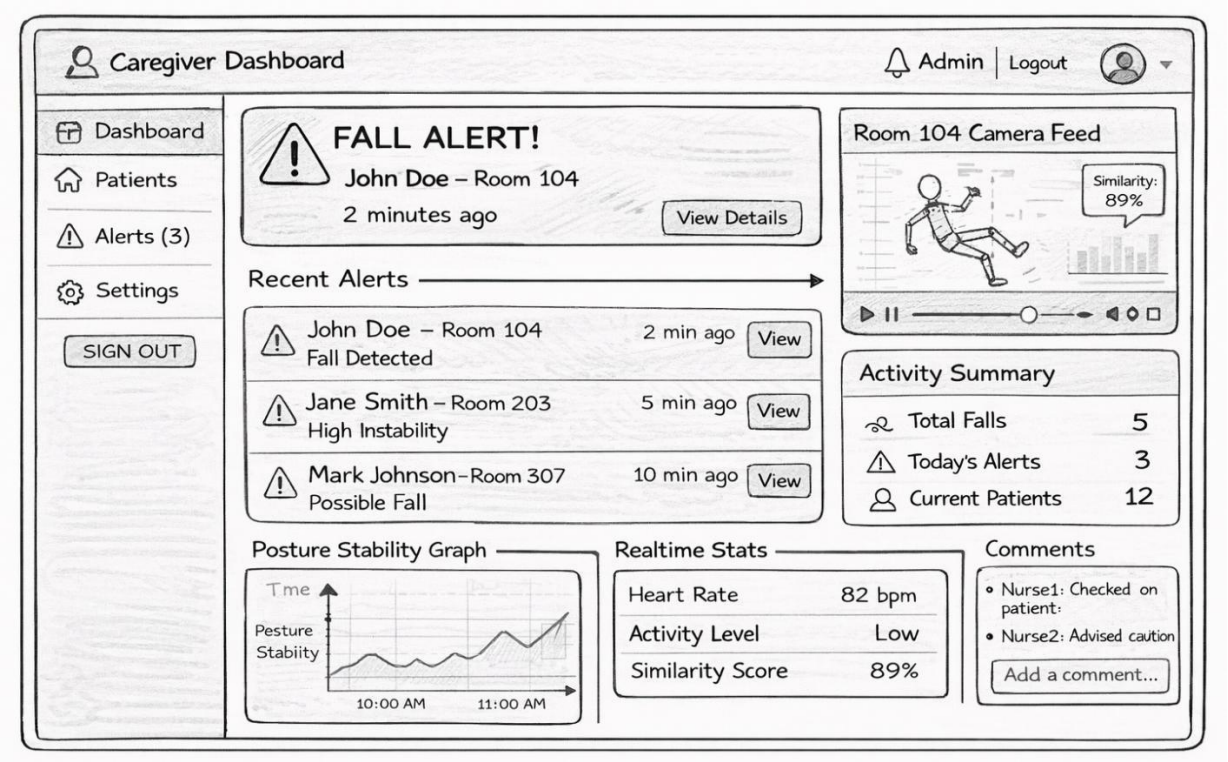
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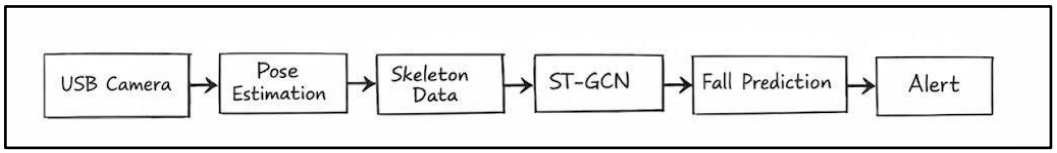
12. APPENDICES

Appendix A: Preliminary UI Mockups



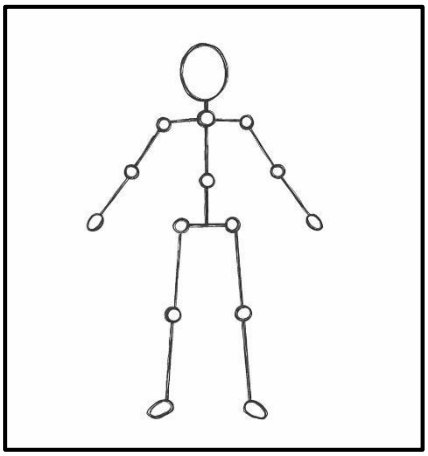
Appendix 1: This section presents the preliminary wireframe design of the caregiver dashboard used for real-time fall monitoring and alert visualization.

Appendix B: System Flow Sketch



Appendix 2 : Conceptual flow of skeletal movement analysis and fall prediction process

Appendix C: Skeleton Representation Sketch



Appendix 3: Example representation of skeletal joint extraction from patient movement